

The Deadly Race

When “graceful recovery” makes coercion lethal

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A formal threat model for wrench attacks — draft, 2026

Not a foreign problem. Not a historical one.

344

documented physical-coercion
attacks

Lopp's

physical-bitcoin-attacks
registry

2024 41

2025 85

2026 48 so far

Brazilian entries include

Florianópolis 2017 · Recife 2021 · São Paulo 2024
Goiânia 2025 · **Vitória, June 2026 · Ribeirão
Preto, July 2026**

12

of them in **Brazil**

The curve is not flattening, and the victims are not billionaires.

Our two standard answers do not work here

- **Cryptography does not help.**

The attacker does not break the cipher. They compel the person who operates it. Dolev–Yao, IND-CCA, forward secrecy — every model we use assumes the adversary cannot reach the *holder*.

- **Obscurity is not security.**

“Never tell anyone you own Bitcoin” is a bet that the attacker does not know something you do. Kerckhoffs settled that in 1883 — and the bet degrades exactly as adoption spreads.

Registry, January 2026, Sweden: a streamer who *had no cryptocurrency at all* was invaded and tortured anyway.

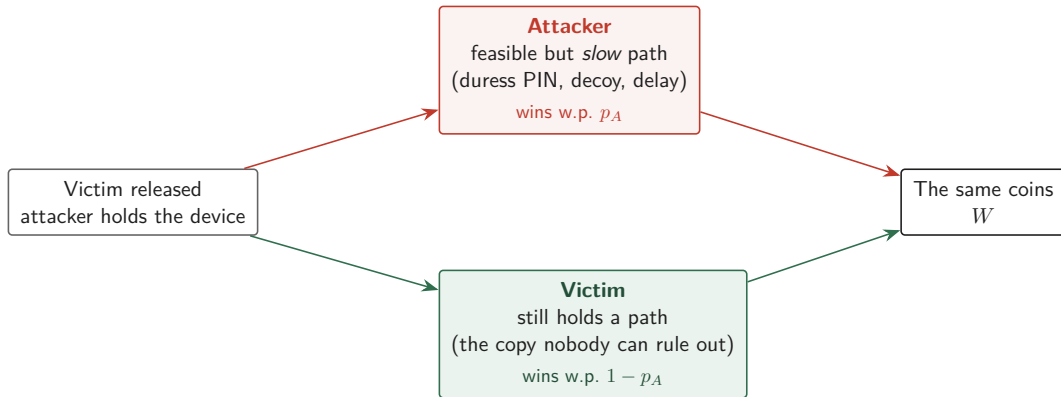
The thing nobody can do

You cannot prove you forgot.

- You can hand over a seed. You can open a wallet. You *cannot* prove there is no second copy — in your head, on an SD card, steganographic notes in a book, the login to an email account.
- Non-possession is a **negative existential**. Cooperation only ever *raises a lower bound* on what you had; it never certifies the upper one.
- **So a rational attacker must assume you kept something** — not from paranoia, but because no evidence could say otherwise.

Non-Negative-Proof Principle (NNPP)

So the encounter ends — and the race starts



Both parties can still reach the funds. That is a **gray outcome**: $0 < p_A < 1$.

What the second racer is worth — dead

Expected take with the victim alive and racing:

$$p_A W$$

Expected take with the victim removed:

$$W$$

$$(1 - p_A) W > c$$

The attacker acts on the difference when it beats the cost c .

And by NNPP, *the only instrument available* for pushing p_A toward 1 is removing the other racer — no promise, no surrender, no cooperation can substitute for it.

Now read the inequality backwards

The *gentler* the fallback, the *smaller* p_A .

The smaller p_A , the **larger** $(1 - p_A)W$.

**Graceful degradation — a virtue everywhere else
in engineering —
is here a **murder incentive**.**

A scheme that simply hands the attacker everything kills nobody. **The graceful fallback is worse than that.**

The designs that market themselves as the safest sit squarely in the region of maximum assassination incentive.

The bar: No Gray Area

$p_A = 0$ fails clean	$0 < p_A < 1$ the race — forbidden	$p_A = 1$ succeeds clean
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- Every encounter must resolve to a **clear** endpoint. **Either you lose clean, or you win clean. Never a race.**
- The two endpoints are not equally *pleasant* — $p_A = 1$ costs you the funds — but both are **survivable**.
- Coercion resistance is a bar on **the holder's safety** first, and only then on the funds.

Exactly two ways to clear it

(R1) Remove the race

No feasible seizure path exists at all
 $\Rightarrow p_A = 0$.

Custody stays **individual**.

The harder road. This is what my construction attempts — and it is *research*, not a product.

(R2) Winner out of reach

The party who *will* win is a remote delegate the attacker cannot touch.

Custody becomes **shared** — you pay with a trusted party.

Timelock-rescue vaults with a remote co-signer. **This ships today.**

If you take one practical thing from tonight: **R2 is the option that already exists.**

The crypto isn't broken. The incentive is.

And it lives in a layer our threat models leave out.

- Paper (draft): github.com/Yuri-SVB/great-wall-docs/papers/wrench-attack-threat-model
- Registry: github.com/jlopp/physical-bitcoin-attacks
- BIP-450 · github.com/Yuri-SVB

It is a draft, and I would rather it be attacked than admired.

Come find me if you want to (a) poke holes in it, (b) tell me about a Brazilian case that is not in the registry — I contribute to it, and I will add it — or (c) just get the link.

Backup slides — Q&A

“Doesn’t multisig solve this?”

- Only if it clears the bar. **Reachable co-signers who can be coerced one at a time is a race with extra steps.**
- R2 works **only** when the deciding party is genuinely beyond the attacker’s reach.
- Honest cost of R2: its safety rests on the delegate *refusing a legitimate owner pleading for their life*. The refusal that defeats the coerced plea also defeats the genuine one.
- And an access-dominating delegate is **not self-custody** — it reinstates the counterparty the design set out to remove.

“Real attackers aren’t rational calculators.”

- The model is deliberately *a fortiori*: it endows the attacker with the *most kill-prone* reasoning available and asks whether the design survives anyway.
- It is a **floor**, not a claim that every attacker computes an expected value.
- Same move for Kerckhoffs and for NNPP: give the adversary everything, then demand safety.

“If, even after maximizing the attacker’s proclivity to enact the deadly race, killing still buys it nothing, the method is deadly-race-safe.”

The open question I have not closed

The R1 route needs a secret that **cannot be transmitted** — one you can use but cannot hand over. That means tacit, perceptual recall (*TKBA*), not a memorised string.

- It is the **load-bearing assumption** of the construction.
- It cannot be settled by argument — it needs **measurement**: how reliably do people re-point a memorised location, over realistic intervals, against an explicit false-reject target?
- Spaced repetition addresses *maintenance* of recall, not the base rate.

That is an open question, not a hole I have papered over — and it is the sharpest critique the work has drawn so far.

“Isn’t publishing this irresponsible?”

- The incentive is **disclosed, not created**. It is already a property of the situation a coercer faces.
- Kerckhoffs again: assume they understand the landscape. Not publishing is a bet on obscurity — and the registry is the record of that bet losing.
- Publishing the danger *together with a bar that rejects the designs causing it* is net-protective.
- **Coldcard’s five quiet years were not obscurity working — they were the exposed population compounding.** Delay never chooses between attack and no attack, only between a smaller reckoning now and a larger one later.

Not peer reviewed. First manuscript, 2026, in preparation — said plainly.